

CRANBROOK
SCHOOL

Year 12 Extension 1 Mathematics

Mini Examination

Wednesday April 8, 2009

Instructions

- There are four (4) questions, each worth 15 marks
- Attempt all questions
- Answer each question in a new booklet
- Show all necessary working
- Calculators are allowed in all sections
- 5 minutes reading time

Time Allowed: 90 minutes

Total Marks: 60

- (a) Consider the function $P(x) = x - \ln 10x$.
- (i) Show that a root exists between $x = 3$ and $x = 4$. 1
- (ii) By choosing $x = 3.6$ as a first approximation and applying Newton's Method once determine a second approximation to this root. 2
- (iii) Comment on the accuracy of your second approximation. 1
- (iv) Why would Newton's Method have failed if $x = 1$ had been chosen as the first approximation? 1
- (b) If α, β and γ are the roots of $x^3 + 4x^2 + 8x + 16 = 0$, find the value of
- (i) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ 2
- (ii) $\alpha^2 + \beta^2 + \gamma^2$ 2
- (c) The polynomial $P(x) = x^5 + 3x^4 - 10x^3 + 2x^2 + 9x - 5$ has a triple root at $x = 1$ and two other single roots. Determine the values of these other roots and express $P(x)$ as a product of its factors. 3
- (d) A polynomial $Q(x) = x^4 + px^3 + qx^2 - 5x + 1$ has a zero at $x = 1$. When $Q(x)$ is divided by $x^2 + 2$ it has a remainder of $1 - 7x$. Find p and q . 3

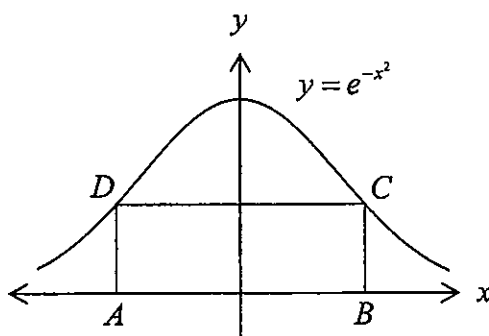
- (a) (i) Use the substitution $u = 1 - x^6$ to find $\int \frac{x^5}{\sqrt{1-x^6}} dx$ 3
- (ii) Use the substitution $u = 1 + \log_e x$ to evaluate $\int_1^e \frac{dx}{x(1 + \log_e x)^2}$ 3
- (b) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$. Normals to this parabola at P and Q meet at the point R .
- (i) Prove that R has coordinates $[-apq(p+q), a(p^2 + pq + q^2 + 2)]$ 4
- (ii) If the normals intersect at right angles prove that the locus of R is the parabola $x^2 = a(y - 3a)$. 4
- (iii) Hence find the coordinates of the focus of the locus of R . 1

(a) Differentiate $y = \ln(x^3 \sqrt{x^2 + 1})$ 2

(b) Evaluate $\int_1^3 \left(2x + \frac{3}{x^2}\right)^2 dx$ 3

(c) Find the exact value of the area enclosed by the curve $y = \frac{e^x}{1+e^x}$, the x -axis, and the lines $x = 0$ and $x = 1$. 3

(d) $ABCD$ is a rectangle drawn between the curve $y = e^{-x^2}$ and the x -axis. 4



(i) Show that $ABCD$ has area $2xe^{-x^2}$ units²

(ii) Hence find the maximum area of such a rectangle.

(e) Write down the derivative of $(x-1)e^x$ and use your result to evaluate $\int_{-1}^1 xe^x dx$ 3

Question 4 (15 Marks)**START A NEW BOOKLET****Marked by HRK**

- (a) Prove by mathematical induction where n is a positive integer,
 $3^{3n} + 2^{n+2}$ is divisible by 5. 6
- (b) For the curve $y = xe^{-x}$, 9
- (i) Determine the stationary point and the point of inflexion.
- (ii) Sketch the curve.
- (iii) From your sketch, show that the equation $xe^{-x} = k$ has
- (α) Two roots if $0 < k < \frac{1}{e}$
- (β) One real root if $k \leq 0$
- (γ) No real roots if $k > \frac{1}{e}$

END OF EXAMINATION

① (a) $P(x) = x - \ln 10x$

(i) $P(3) = 3 - \ln 30 = -0.40119\dots$

$P(4) = 4 - \ln 40 = 0.31112\dots$

As $P(3)$ and $P(4)$ have different signs and $P(x)$ is continuous for $x > 0$ $\therefore$ at least 1 root exists in interval $3 < x < 4$.

(ii) Let $z_1 = 3.6$

$P(x) = x - \ln 10x$

$\therefore P'(x) = 1 - \frac{1}{x}$

By Newton's Method

$z_2 = z_1 - \frac{P(z_1)}{P'(z_1)}$

$\therefore z_2 = 3.6 - \frac{P(3.6)}{P'(3.6)}$
 $= 3.6 - \frac{-0.01648\dots}{0.7222\dots}$

$= 3.577180069\dots$

(iii) Now $P(z_2) = 0.000020175\dots$
 $\approx 2.0 \times 10^{-5}$

$\therefore$ As its order of accuracy has a magnitude of 10^{-5} its accuracy is very good.

(iv) If $x=1$ $P'(1) = 0$. This would have meant that z_2 would have been undefined and Newton's Method would have failed. Indeed at $x=1$ a stationary point exists and any tangent drawn to this point would not have cut the x -axis meaning that Newton's Method would not have applied for finding a closer approximation to the root.

(b) $x^3 + 4x^2 + 8x + 16 = 0$ has roots α , β and γ

Now $\alpha + \beta + \gamma = -\frac{b}{a} = -4$

$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = 8$

$\alpha\beta\gamma = -\frac{d}{a} = -16$

(i) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$

$= \frac{8}{-16}$

$= -\frac{1}{2}$

(ii) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$
 $= (-4)^2 - 2(8)$
 $= 16 - 16$
 $= 0$

(c) $P(x) = x^5 + 3x^4 - 10x^3 + 2x^2 + 9x - 5$

As $P(x)$ has a triple root at $x=1 \Rightarrow (x-1)^3$ is a factor.

Now possible zeros of $P(x)$ are $\pm 1, \pm 5$.

Let $x = -1 \therefore P(-1) = -1 + 3 + 10 + 2 - 9 - 5 = 0$

$\therefore x+1$ is a further factor

Let $x = -5 \therefore P(-5) = -3125 + 1875 + 1250 + 50 - 45 - 5 = 0$

$\therefore x+5$ is the other factor

$\Rightarrow$ other roots of $P(x)$ are $x = -1$ and $x = -5$.

$\therefore P(x) = (x-1)^3(x+1)(x+5)$

$$(d) \quad Q(x) = x^4 + px^3 + qx^2 - 5x + 1$$

As $Q(x)$ has a zero at $x=1 \quad \therefore Q(1)=0$

$$\therefore 0 = 1 + p + q - 5 + 1$$

$$\therefore p + q = 3 \quad \text{--- (1)}$$

$$\begin{array}{r}
 x^2 + 2 \overline{) x^4 + px^3 + qx^2 - 5x + 1} \\
 \underline{-(x^4 + 2x^2)} \\
 px^3 + x^2(q-2) - 5x + 1 \\
 \underline{-(px^3 + 2px)} \\
 x^2(q-2) + x(-5-2p) + 1 \\
 \underline{-(x^2(q-2) + 2(q-2))} \\
 x(-5-2p) + (5-2q)
 \end{array}$$

But the remainder is $-7x + 1$.

$$\therefore -7 = -5 - 2p \quad \text{--- (2)}$$

$$1 = 5 - 2q \quad \text{--- (3)}$$

$$\left. \begin{array}{l} \text{from (2) } 2p = 2 \therefore p = 1 \\ \text{and from (3) } 2q = 4 \therefore q = 2 \end{array} \right\} \text{ which satisfies (1)}$$

$$\therefore (p, q) = (1, 2).$$

$$2(a) (i) \quad I = \int \frac{x^5}{\sqrt{1-x^6}} dx$$

$$\text{Let } u = 1 - x^6$$

$$\therefore \frac{du}{dx} = -6x^5$$

$$\therefore \frac{du}{-6} = x^5 dx$$

$$\begin{aligned} \therefore I &= \int \frac{du}{-6} \cdot \frac{1}{\sqrt{u}} \\ &= -\frac{1}{6} \int u^{-\frac{1}{2}} du \\ &= -\frac{1}{6} \frac{u^{\frac{1}{2}}}{\frac{1}{2}} + c \\ &= -\frac{1}{3} \sqrt{1-x^6} + c \end{aligned}$$

$$(ii) \quad I = \int_1^e \frac{dx}{x(1+\log_e x)^2}$$

$$\text{Let } u = 1 + \log_e x$$

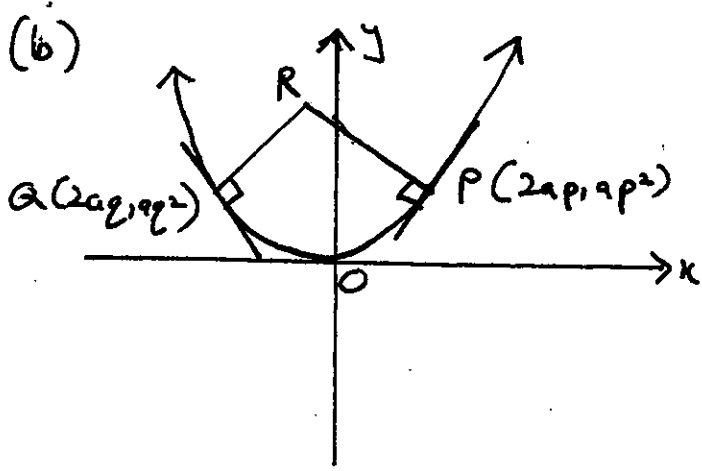
$$\therefore \frac{du}{dx} = \frac{1}{x}$$

$$\therefore du = \frac{dx}{x}$$

$$\text{when } x=1 \quad u=1$$

$$x=e \quad u=2$$

$$\begin{aligned} \therefore I &= \int_1^2 \frac{du}{u^2} \\ &= \left[\frac{u^{-1}}{-1} \right]_1^2 \\ &= - \left[\frac{1}{u} \right]_1^2 \\ &= - \left[\frac{1}{2} - 1 \right] \\ &= \frac{1}{2} \end{aligned}$$



(i) $x^2 = 4ay \therefore y = \frac{x^2}{4a}$
 $\therefore \frac{dy}{dx} = \frac{2x}{4a} = \frac{x}{2a}$
 At $P(2ap, ap^2)$ $\frac{dy}{dx} = \frac{2ap}{2a} = p$
 $\therefore \text{slope} = p \therefore m_{\text{norm}} = -\frac{1}{p}$

$\therefore$ Eqn of normal at P is:

$$y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$\therefore py - ap^3 = -x + 2ap$$

$$\therefore x + py = 2ap + ap^3 \quad \text{--- (1)}$$

Similarly eqn of normal at Q is:

$$x + qy = 2aq + aq^3 \quad \text{--- (2)}$$

$$\textcircled{1} - \textcircled{2}: y(p-q) = 2a(p-q) + a(p^3 - q^3)$$

$$\therefore y = \frac{2a(p-q) + a(p-q)(p^2 + pq + q^2)}{p-q}$$

$$\therefore y = 2a + a(p^2 + pq + q^2)$$

$$\therefore y = a(p^2 + pq + q^2 + 2) \text{ sub into (1)}$$

$$\therefore x + pa(p^2 + pq + q^2 + 2) = 2ap + ap^3$$

$$\therefore x + \cancel{ap^3} + ap^2q + \cancel{apq^2} + 2ap = \cancel{2ap} + \cancel{ap^3}$$

$$\therefore x = -apq(p+q)$$

$$\Rightarrow R = [-apq(p+q), a(p^2 + pq + q^2 + 2)]$$

(ii) As normals meet at $90^\circ \Rightarrow m_{\text{norm } p} \cdot m_{\text{norm } q} = -1$

$$\therefore -\frac{1}{p} \cdot -\frac{1}{q} = -1 \Rightarrow pq = -1$$

$$\therefore R = [a(p+q), a(p^2 + q^2 + 1)]$$

$$\therefore x = a(p+q) \therefore p+q = \frac{x}{a} \quad \text{--- (1)} \quad \text{and } y = a(p^2 + q^2 + 1) \therefore \frac{y}{a} - 1 = p^2 + q^2 \quad \text{--- (2)}$$

But $p^2 + q^2 = (p+q)^2 - 2pq \therefore p^2 + q^2 = (p+q)^2 + 2$

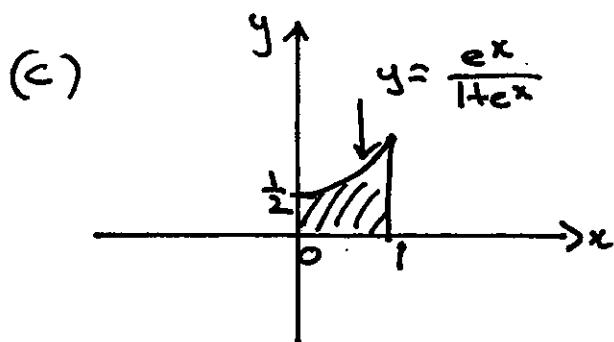
$$\Rightarrow \frac{y}{a} - 1 = \left(\frac{x}{a}\right)^2 + 2 \therefore \frac{y}{a} - 3 = \frac{x^2}{a^2} \therefore x^2 = a(y - 3a)$$

(iii) From (i) $x^2 = a(y - 3a)$
 is of the form $x^2 = 4A(y - k)$
 $\therefore \text{vertex} = (0, 3a)$
 $4A = a \therefore A = \frac{a}{4}$
 $\therefore \text{focal length} = \frac{a}{4}$
 $\Rightarrow \text{focus of the locus of R}$
 is: $(0, \frac{13a}{4})$

$$\begin{aligned}
 \text{Q3 (a) Let } y &= \ln(x^3 \sqrt{x^2+1}) \\
 &= \ln x^3 + \ln (x^2+1)^{\frac{1}{2}} \\
 &= 3 \ln x + \frac{1}{2} \ln(x^2+1)
 \end{aligned}$$

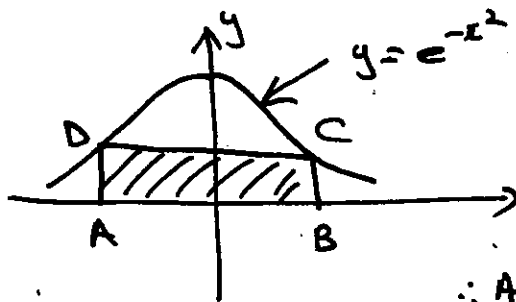
$$\begin{aligned}
 \therefore \frac{dy}{dx} &= \frac{3}{x} + \frac{2x}{2(x^2+1)} \\
 &= \frac{3}{x} + \frac{x}{x^2+1} \\
 &= \frac{3x^2+3+x^2}{x(x^2+1)} \\
 &= \frac{4x^2+3}{x(x^2+1)}
 \end{aligned}$$

$$\begin{aligned}
 \text{(b) } I &= \int_1^3 \left(2x + \frac{3}{x^2}\right)^2 dx \\
 &= \int_1^3 \left(4x^2 + \frac{12}{x} + 9x^{-4}\right) dx \\
 &= \left[\frac{4x^3}{3} + 12 \ln x - 3x^{-3} \right]_1^3 \\
 &= \left[\left(36 + 12 \ln 3 - \frac{1}{9}\right) - \left(\frac{4}{3} + 0 - 3\right) \right] \\
 &= \frac{338}{9} + 12 \ln 3
 \end{aligned}$$



$$\begin{aligned}
 \therefore \text{Area} &= \int_0^1 \frac{e^x}{1+e^x} dx \\
 &= \left[\ln(1+e^x) \right]_0^1 \\
 &= \left[\ln(1+e) - \ln(1+1) \right] \\
 &= \ln\left(\frac{1+e}{2}\right) \text{ units}^2
 \end{aligned}$$

(d) (i)



Let $B = (x, 0) \therefore A = (-x, 0)$
 $C = (x, e^{-x^2}), D = (-x, e^{-x^2})$

$$\begin{aligned} \therefore \text{Area of } ABCD &= LB \\ &= 2x \times e^{-x^2} \\ &= 2x e^{-x^2} \text{ units}^2 \end{aligned}$$

(ii)

$$\begin{aligned} A &= 2x e^{-x^2} \\ \therefore A' &= 2x \cdot e^{-x^2} \cdot (-2x) + e^{-x^2} \cdot 2 \\ &= 2e^{-x^2} [-2x^2 + 1] \end{aligned}$$

For a possible max/min $A' = 0 \quad \therefore 1 = 2x^2 \quad (e^{-x^2} > 0)$
 $\therefore x = \frac{1}{\sqrt{2}} \quad (x > 0)$

x	$\frac{1}{\sqrt{2}}^-$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}^+$
A'	+	0	-

$\Rightarrow$ max. area when $x = \frac{1}{\sqrt{2}}$

$$\begin{aligned} \therefore \text{Max. area} &= \frac{2}{\sqrt{2}} e^{-\frac{1}{2}} \\ &= \sqrt{2} e^{-\frac{1}{2}} \text{ units}^2 \end{aligned}$$

(e)

Let $y = (x-1)e^x$
 $\therefore y' = (x-1) \cdot e^x + e^x \cdot 1$
 $= e^x (x-1+1)$
 $= x e^x$

$$\begin{aligned} \text{Now } \int_{-1}^1 x e^x dx &= \left[(x-1)e^x \right]_{-1}^1 \\ &= [0 - (-2)e^{-1}] \\ &= \frac{2}{e} \end{aligned}$$

4(a) TO PROVE: $3^{3n} + 2^{n+2}$ is divisible by 5 for $n \in \mathbb{N}^+$

PROOF: Step 1: When $n=1$ $3^{3n} + 2^{n+2} = 3^3 + 2^3$
 $= 27 + 8$
 $= 35$

which is divisible by 5

$\therefore$ it is true for $n=1$.

Step 2: Assume it is true for $n=k$ ($k \in \mathbb{N}$, $k \in \mathbb{N}^+$) and prove it is true for $n=k+1$.

i.e. $3^{3k} + 2^{k+2} = 5M$ (where $M \in \mathbb{N}$)

$$\therefore 3^{3k} = 5M - 2^{k+2} \quad \text{--- (1)}$$

$$\begin{aligned} \text{When } n=k+1 \quad 3^{3n} + 2^{n+2} &= 3^{3(k+1)} + 2^{k+1+2} \\ &= 3^{3k} \cdot 3^3 + 2^{k+3} \\ &= (5M - 2^{k+2}) \cdot 27 + 2^{k+3} \quad (\text{sub (1)}) \\ &= 135M - 27 \cdot 2^{k+2} + 2^{k+2} \cdot 2 \\ &= 135M - 25 \cdot 2^{k+2} \\ &= 5[27M - 5 \cdot 2^{k+2}] \end{aligned}$$

which is divisible by 5

$\therefore$ if it is true for $n=k$ so it is true for $n=k+1$.

Step 3: It is true for $n=1$ and so it is true for $n=1+1=2$.
It is true for $n=2$ and so it is true for $n=2+1=3$
and so on for all positive integral values of n .

(b)

$$y = x e^{-x}$$

(i)

$$y' = x \cdot -e^{-x} + e^{-x} \cdot 1$$

$$= -e^{-x} [x-1]$$

$$y'' = -e^{-x} \cdot 1 + (x-1) \cdot -e^{-x}$$

$$= -e^{-x} [1 - (x-1)]$$

$$= -e^{-x} [2-x]$$

For a stat. pt $y' = 0 \quad \therefore x=1 \quad (e^{-x} > 0)$

When $x=1 \quad y'' < 0 \Rightarrow$ max. turn. pt at $(1, \frac{1}{e})$

For a possible pt of inflexion $y'' = 0 \quad \therefore x=2 \quad (e^{-x} > 0)$

x	2^-	2	2^+
y''	$-$	0	$+$

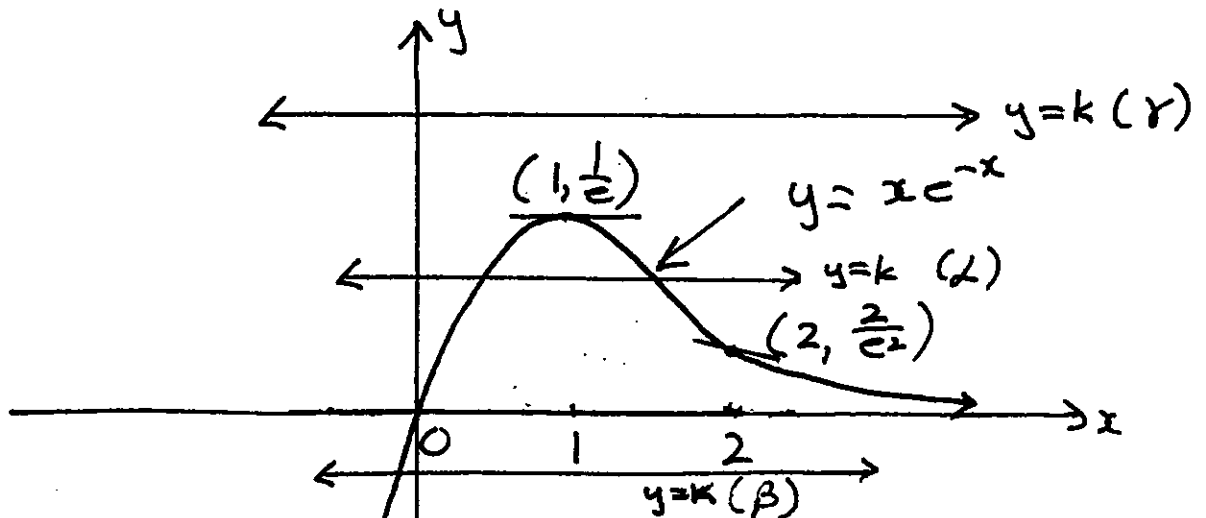
concavity change $\Rightarrow$ pt. of inflexion at $(2, \frac{2}{e^2})$

As $x \rightarrow \infty \quad y \rightarrow 0$

As $x \rightarrow -\infty \quad y \rightarrow -\infty$

Intercept at $(0,0)$

(ii)



The solution to $x e^{-x} = k$ on the graph of $y = x e^{-x}$.

(2) If $0 < k < \frac{1}{e}$

(β) If $k \leq 0$

(γ) If $k > \frac{1}{e}$

can be solved graphically by drawing $y = k$

$y = k$ will cut twice $y = x e^{-x} \therefore 2$ roots

$y = k$ will only cut once $y = x e^{-x} \therefore 1$ real root

$y = k$ will lie above the curve $y = x e^{-x} \therefore$ No real roots